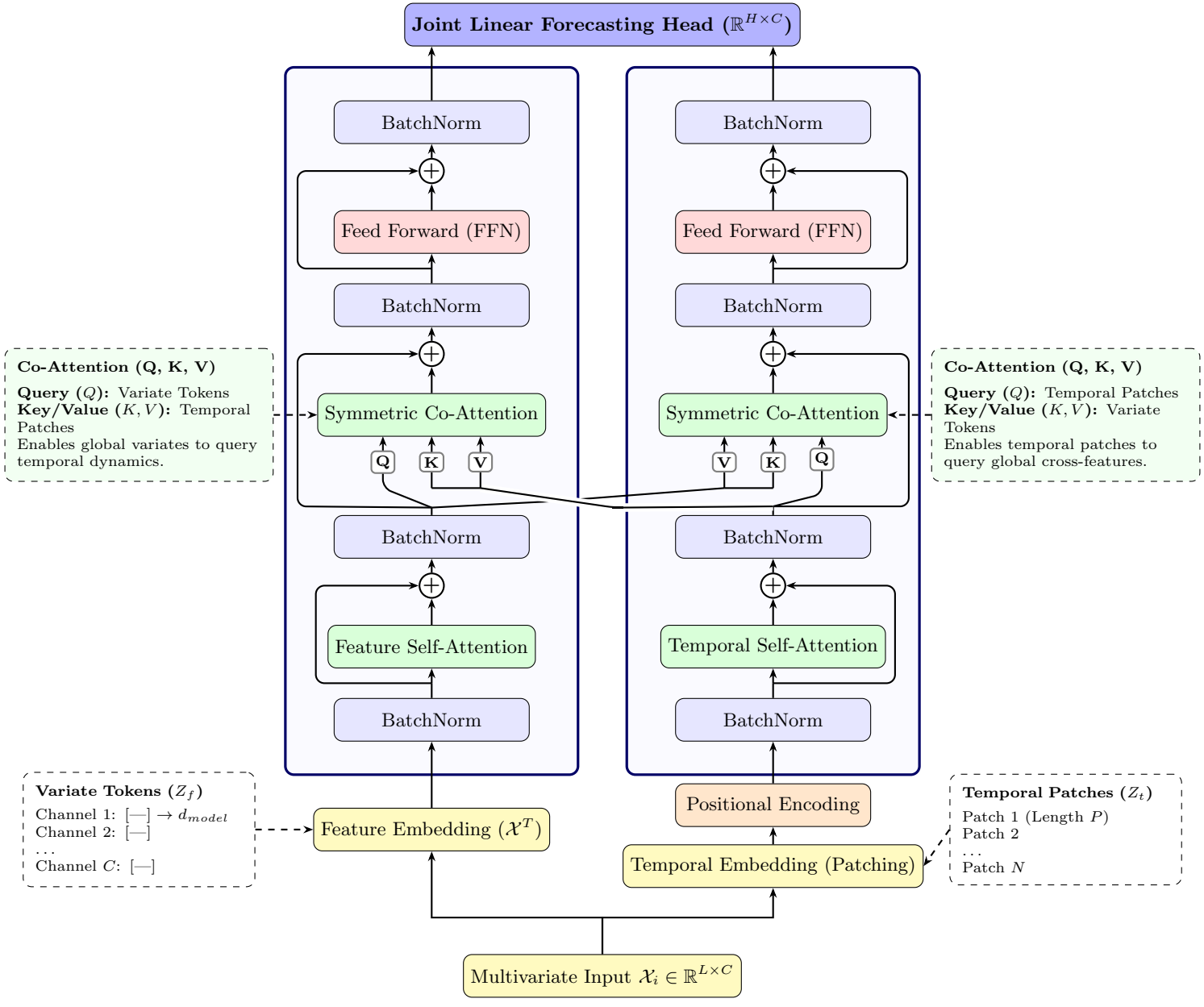

Joint Linear Forecasting Head ($\mathbb{R}^{H\times C}$)
BatchNorm
Feed Forward (FFN)
BatchNorm
Symmetric Co-Attention
Q
K
V
BatchNorm
Feature Self-Attention
BatchNorm
BatchNorm
Feed Forward (FFN)
BatchNorm
Symmetric Co-Attention
V
K
Q
BatchNorm
Temporal Self-Attention
BatchNorm
Co-Attention (Q, K, V)
Query (Q): Variate Tokens
Key/Value (K, V): Temporal Patches
Enables global variates to query temporal dynamics.
Co-Attention (Q, K, V)
Query (Q): Temporal Patches
Key/Value (K, V): Variate Tokens
Enables temporal patches to query global cross-features.
Variate Tokens (Z_f)
Channel 1: [—] → d_{model}
Channel 2: [—]
...
Channel C: [—]
Feature Embedding ($\mathcal{X}^T$)
Positional Encoding
Temporal Embedding (Patching)
Temporal Patches (Z_t)
Patch 1 (Length P)
Patch 2
...
Patch N
Multivariate Input $\mathcal{X}_i \in \mathbb{R}^{L\times C}$